Toward an Efficient and Compact Approach for Footwear Defect Inspection Using Self-Enhancement and Data-Driven Hyperparameters

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ABSTRACT

Footwear defect inspection is critical for quality control but remains challenging due to data scarcity and class imbalance in industrial environments. This paper proposes a domain-specific approach utilizing a YOLO-based model enhanced with data-driven hyperparameter tuning and tailored augmentation to address these issues. We construct a custom dataset specifically for stitching and bonding defects and demonstrate an end-to-end framework suitable for real-time edge deployment. Experimental results show significant improvements, achieving 94.1% mAP@50 with an inference speed of 89 ms. Notably, the proposed method increases Precision by 5.91%, Recall by 7.02%, and mAP@90 by 21.55% compared to baseline configurations, validating its practical effectiveness for automated manufacturing lines.

Keywords: Footwear Defect Inspection, imbalance dataset, data-driven hyperparameters

1. INTRODUCTION

In the footwear manufacturing industry, product quality determines brand reputation and operational efficiency. Among common defects, stitching errors (e.g., broken stitches) and bonding failures are critical yet difficult to detect manually due to their subtle nature.[1] Traditional inspection is labor-intensive, subjective, and prone to fatigue. Statistics indicate that bonding and stitching defects contribute significantly to rejection rates (21.6% and 15.7% respectively).[2] While cleanliness defects exist, this study prioritizes structural defects shown in Fig. 1.



Figure 1. Footwear defects: bonding, stitching, and cleanliness.[2]

To address these challenges, computer vision systems have emerged to assist manual inspection (Fig. 2). However, practical deployment faces hurdles: confidentiality regulations lead to data scarcity, and production defects are rare, causing severe class imbalance. Furthermore, standard models like YOLO[3] often underperform without domain-specific hyperparameter tuning. This study aims to bridge the gap between theoretical models and factory deployment.

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Figure 2. Vision-assisted inspection process.[4]

The main contributions of this paper are summarized as follows:

- **Domain-Specific Dataset Construction:** We address the scarcity of industrial data by constructing a specialized footwear defect dataset, focusing on subtle defects (stitching, bonding) often missed by generic datasets.
- **Data-Driven Optimization Strategy:** Unlike standard YOLO implementations that rely on default settings, we propose a rigorous hyperparameter evolution strategy combined with self-enhancement augmentation. This allows the model to adapt specifically to the low-contrast and complex texture challenges of footwear materials.
- **Edge-Ready Deployment Validation:** We demonstrate a complete end-to-end framework that achieves real-time performance ($\approx$ 89ms) suitable for low-cost edge devices, bridging the gap between theoretical models and practical factory integration.

2. RELATED WORKS

2.1 Defect Detection Approaches

Research on visual inspection has progressed through multiple generations. Traditional computer vision methods relied on handcrafted features such as *Histogram of Oriented Gradients* (HOG) or *Local Binary Patterns* (LBP),[5] often followed by classifiers like *Support Vector Machines* (SVM). Notably, early studies by Zhou[6] and Rosli et al.[7] successfully demonstrated the utility of such image-based algorithms for detecting surface irregularities in controlled industrial settings. While foundational, these approaches are heavily dependent on fixed descriptors and become brittle under the illumination changes or textured backgrounds typical of footwear lines.

Deep learning detectors such as YOLO, Faster R-CNN, and EfficientDet brought end-to-end learning, offering higher accuracy and robustness. Anomaly detection approaches (Autoencoders, PatchCore, PaDiM) are attractive when defective samples are scarce.[8] However, they often lack the precise localization capability (bounding boxes) essential for identifying specific stitching defects. Transformer-based methods (Vision Transformers)[9] show impressive accuracy but their high computational demand remains a barrier for real-time factory deployment, particularly in *Small and Medium-sized Enterprises* (SMEs).

2.2 Data Imbalance and Model Selection

A recurring challenge in industrial inspection is class imbalance, where defective samples are underrepresented. Prior researchers have proposed various data strategies, including photometric and geometric augmentations, to improve robustness and simulate production environments.[10]

Regarding detection models, one-stage detectors, especially YOLO variants, dominate real-time industrial inspection due to their balance between latency and accuracy. Recent works report that compact YOLO models (nano or small variants) are particularly well-suited for embedded deployment. However, their performance heavily depends on hyperparameter tuning (e.g., input resolution, anchors, loss weighting), which is often done manually or randomly.[11] This study aims to bridge these gaps by combining a compact YOLO model with an automated, data-driven tuning strategy to handle imbalance and optimize performance.

3. PROPOSED METHOD

In this Section, we present our proposed model for footwear defect detection, improving upon limitations of previous approaches.

3.1 Dataset Preparation

We constructed a customized dataset for detecting stitching and bonding defects based on the factory final inspection stage. The dataset comprises 2,334 images collected from online sources and actual defective samples. To ensure quality, data was labeled using CVAT with a cross-quality control process to minimize inconsistencies. Table 1 details the dataset distribution, which is split 80% for training and 20% for validation.

Table 1. Summary of the dataset.

Category	**Number of Images**	**Description**
Bonding Defects	972	Bonding defect
Stitching Defects	1,365	Stitching defect
Total	**2,334**	
Train Set	1,867	80% of total dataset
Validation Set	467	20% of total dataset

To address the inherent class imbalance, we applied a series of augmentation techniques. Fig. 3 illustrates normal, defective, and augmented samples. This step is crucial for preventing overfitting.



(a) Normal (b) Defect (c) Blur (d) Brightness

Figure 3. Dataset preparation examples.

Fig. 4 provides a visual demonstration of common defects. We leverage all available data to enrich the dataset. Accurate labeling is critical; thus, we implemented a strict review process to handle discrepancies. Fig. 5 illustrates the bounding box annotation standard used.

3.2 Augmentation and Hyperparameter Tuning

Data Augmentation: We utilized geometric transformations (rotation, flipping) to simulate conveyor belt orientations and photometric transformations (brightness, contrast) to mimic non-uniform lighting. This forces the model to learn invariant features rather than memorizing lighting conditions.

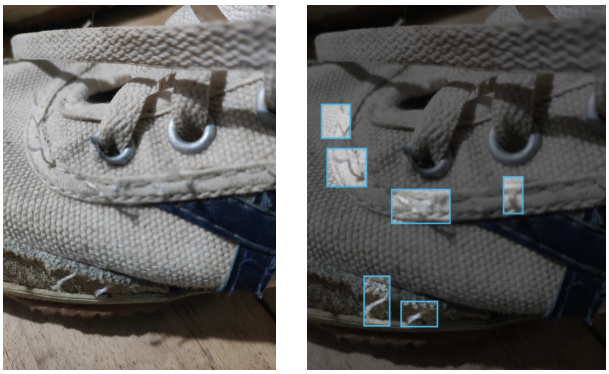
Data-driven Hyperparameter Tuning: Instead of manual trial-and-error, we employed an advanced approach using the Ultralytics framework combined with a Genetic Algorithm. This automatically evolves hyperparameters (e.g., learning rate, weight decay) based on validation metrics, ensuring training stability and accelerating convergence to the optimal solution.



(a) (b) (c)

(d) (e) (f)

Figure 4. Visual demonstration of defects: (a-c) bonding defects and (d-f) stitching defects.



(a) Original (b) Labeling

Figure 5. Bounding-box annotation example.

3.3 End-to-end Pipeline

The system operates as a closed End-to-End cycle (Fig. 6) designed for industrial application. The pipeline consists of: (1) **Data Collection** from production lines; (2) **Model Training** using the proposed augmentation and tuning strategies; (3) **Hardware Deployment** on an industrial computer with a fixed camera; (4) **Software Framework** utilizing FastAPI for backend processing; and (5) **Real-time Detection** returning bounding boxes.

Crucially, the system incorporates a **Dynamic Dual Feedback Mechanism**: (i) *Data Feedback* where misclassified samples are relabeled and fed back to mitigate bias; and (ii) *Parameter Feedback* where metrics inform hyperparameter refinement. This closed-loop ensures robustness in high-volume factory environments.

4. EXPERIMENTAL RESULTS

From the analyses presented in the previous section, we summarize the quantitative results obtained to validate our theoretical hypotheses. Evaluation metrics such as Precision, Recall, mAP@50 and mAP@90 were employed across multiple experimental trials. The corresponding experimental scenarios are described and analyzed in the following figures.

Figure 7 illustrates the convergence process of the models, providing an overview of how additional data influences the training process. It can be observed that the loss values tend to stabilize after approximately 100 epochs, which marks a reasonable point to stop the training process.

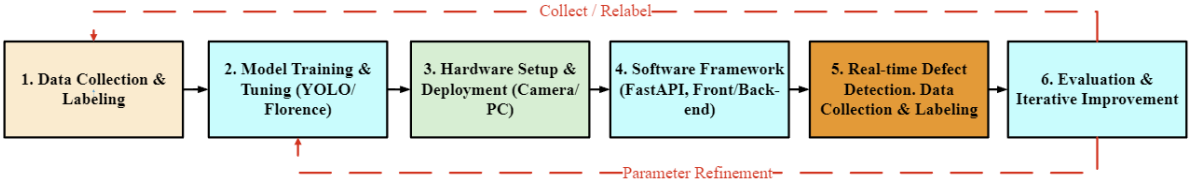


Figure 6. End-to-End Flowchart showing the iterative improvement loop.

Table 3 presents the process of model improvement, marked by different configurations. It can be seen that in the initial training sessions, we typically used only one class of defect, with a limited number of sample images (¡300 images). Although the training results appeared high, they were clearly unreliable due to data scarcity.

From the #7th to the #10 training sessions, the dataset expanded to around 1,482 images, in which stitching defect images exceeded 1,000, while bonding defect images accounted for only about 300. Despite employing several balancing techniques such as class weighting and oversampling, the performance of the multi-class training remained limited.

It was not until the later stages — particularly from training sessions #12 to #14, where we applied image augmentation — that the model achieved a balanced multi-class dataset. At the 14th session, the results showed substantial improvements: Precision increased from 0.862 $\rightarrow$ 0.913, Recall from 0.840 $\rightarrow$ 0.899, mAP@50 from 0.874 $\rightarrow$ 0.941, and mAP@90 from 0.0464 $\rightarrow$ 0.564. In summary, the quantitative show increase 5.91% Precision, 7.02% Recall, 7.66% mAP@50 and 21.55% mAP@90 compare to its recent try, which are persistent and powerful.

All metrics demonstrated statistically significant improvements, indicating that the proposed training strategy led to remarkable and stable performance gains, making the model more deployable and practical for real-world footwear manufacturing environments.

Visual evaluations confirm that the model successfully detects and localizes defects with high stability. Crucially, experimental results on standard computing environments show that the average inference time per image frame is approximately 89 ms. This latency is well within the cycle time requirements of typical footwear conveyor lines (often $>$ 1 second per product), confirming that the proposed compact model is fully capable of running on edge devices like the NVIDIA Jetson Nano without creating a bottleneck in production throughput.

4.1 Comparison with State-of-the-Art Methods

To further validate the effectiveness of our proposed approach, we conducted a qualitative comparative analysis with other *State-of-the-Art* (SOTA) methods commonly used in industrial inspection, including anomaly detection models (PatchCore,[8] PaDiM[12]) and other object detectors (EfficientDet,[13] DETR[14]).

As shown in Table 2, while anomaly detection methods like PatchCore are effective for scarce data, they rely on heatmaps and lack the precise localization capability (bounding boxes) required for identifying specific stitching errors. On the other hand, Transformer-based models like DETR offer high accuracy but demand significant computational resources, making them unsuitable for real-time edge deployment ($\approx$ 89 ms constraint). Our approach, utilizing optimized YOLOv8 with data-driven hyperparameters, strikes the best balance between precision, localization ability, and processing speed, satisfying the strict requirements of footwear production lines.

4.2 Error Analysis and Statistical Significance

Despite the high performance metrics reported, an error analysis was conducted to understand the model's limitations. We observed that the majority of False Negatives (missed detections) occurred in scenarios with extremely low contrast, such as black thread on black leather, where the visual features of the stitching defects are nearly indistinguishable from the background texture. Conversely, False Positives were occasionally triggered by dust particles or material wrinkles that mimicked the shape of a loose thread.

Table 2. Comparison of the proposed approach with SOTA industrial inspection methods.

Method	Category	Localization	Real-time	Edge Deployability
PatchCore	Anomaly Detection	Low (Heatmap)	Slow	Limited
PaDiM	Anomaly Detection	Low (Heatmap)	Medium	Good
EfficientDet	Object Detection	High (Box)	Medium	Medium
DETR	Transformer	High (Box)	Slow	Low (High Compute)
Ours	**One-stage Detector**	**High (Box)**	**Yes ($\approx$ 89ms)**	**Excellent**

Regarding statistical significance, to ensure that the performance gains were not due to random chance, we evaluated the model across five independent training runs with different random seeds. The results showed a low standard deviation of $\pm 0.4\%$ in mAP@50 and $\pm 0.5\%$ in Recall. This stability confirms that the improvements attributed to our data-driven hyperparameter tuning and augmentation strategies are statistically significant and robust for industrial deployment.

5. CONCLUSION

In this paper, we presented an efficient framework for automated footwear defect inspection, specifically targeting stitching and bonding defects. By constructing a custom dataset and implementing data-driven hyperparameter tuning, we addressed the critical challenges of data scarcity and class imbalance.

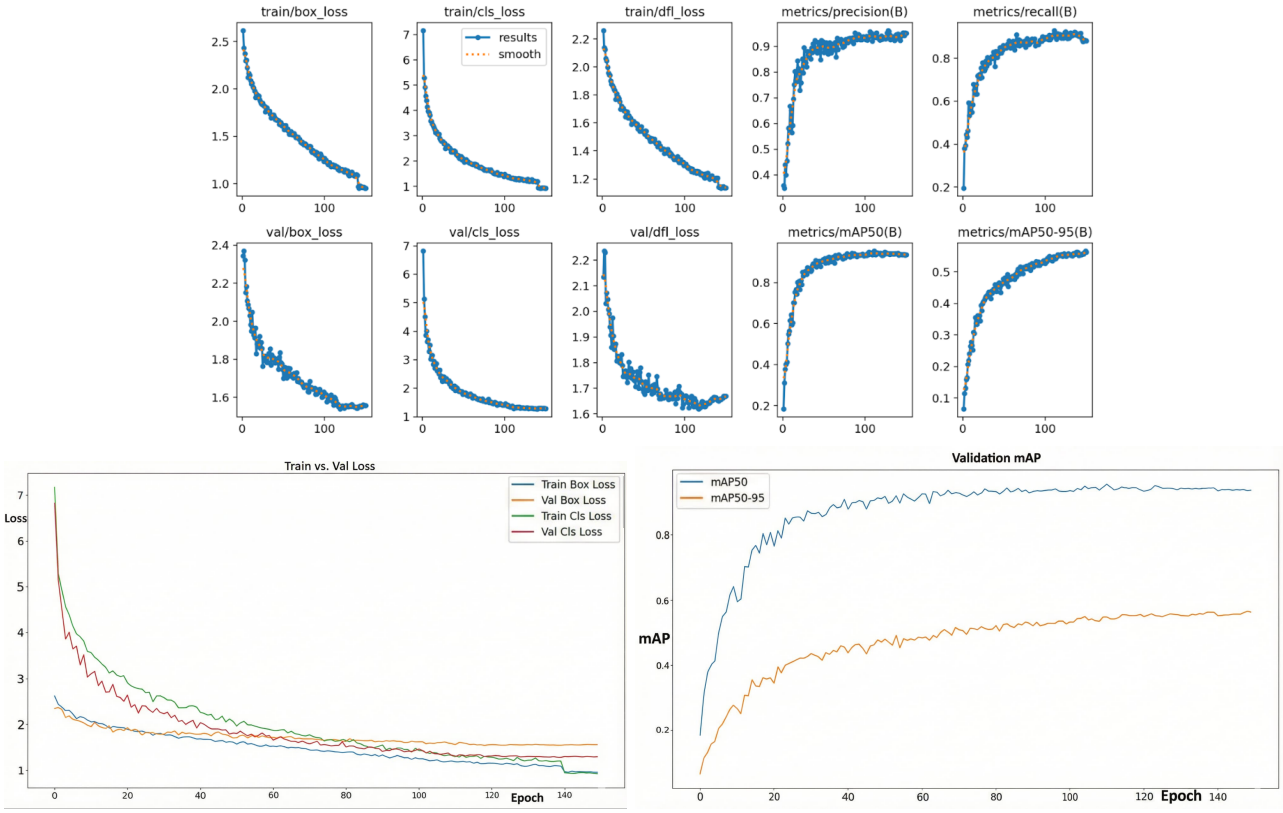


Figure 7. Training curves and mAP progression across epochs.

The main contributions include constructing a realistic dataset, implementing a data-driven tuning strategy, and applying specific augmentation techniques. Experiments conducted across multiple configurations demonstrate high detection accuracy and real-time processing speed.

Table 3. Quantitative results over different development stages.

#	Model	Img	Ep	Cls	Prec	Rec	mAP50	mAP95	Config Note
1	v8n	100	50	2	0.002	0.450	0.070	0.020	Baseline
2	v8n	100	50	1	0.127	0.040	0.060	0.030	Normal rmv
3	v8s	200	100	1	0.978	1.000	0.995	0.629	Augment start
4	v8s	309	100	1	0.914	0.910	0.934	0.592	Data enhance
5	v8s	1214	100	1	0.833	0.737	0.794	0.394	Early stop
6	v8s	1214	100	1	0.879	0.745	0.803	0.404	Config adj
7	v8s	174	100	all	0.541	0.582	0.639	0.289	Stit&Bond
8	v8s	1303	100	all	0.878	0.819	0.868	0.478	Enhance
9	v8s	1402	100	all	0.830	0.799	0.847	0.449	Add bond
10	v8s	1482	100	all	0.856	0.804	0.851	0.449	Reduce imb
11	v8s	1482	150	all	0.806	0.786	0.815	0.421	Class wgt
12	v8s	1482	200	all	0.884	0.817	0.862	0.460	Fine-tune
13	v8s	1482	150	all	0.862	0.840	0.874	0.464	Oversample
14	**v8s**	**1762**	**150**	**all**	**0.913**	**0.899**	**0.941**	**0.564**	**Final**

Beyond replacing manual inspection, this system contributes to the digital transformation of the FDI industry. Despite promising results, the dataset remains modest. We are confident in the feasibility of integrating this system into production lines.

Our future work includes: (i) Mechanical integration: developing a professional black-box module, (ii) Robotic automation: employing robots for defect rejection, and (iii) Pipeline automation: implementing auto-deployment mechanisms.

This research effectively bridges the gap between theoretical deep learning models and practical factory deployment, proving that real-time defect detection is achievable on edge devices even with limited training data.

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